2.1.2 ▶ Root Mean Square Velocity

The root mean square (RMS) velocity is defined as the square root of the average of the squares of the velocities of the particles in a system. Mathematically (and possibly much more simply),

$$v_{RMS} = \sqrt{\langle v^2 \rangle}. \tag{2.7}$$

RMS quantities appear a lot in statistical physics, and they are used preferential to normal averages for a couple reasons:

1: The average value of an oscillating function may be zero, independent of the amplitude of the oscillations. However, it's hardly intuitive for a wildly-oscillating function to be described the same way as a barely-changing function. The RMS value fixes this discrepancy since it squares all values before taking an average, providing a more accurate representation of a quantity.

2: As we see in 2.1.3, since kinetic energy is defined with the square of the velocity, average velocity is useless if we want to calculate the average kinetic energy. Thus, v_{RMS} is used instead of the normal average.

In any case, since we know the PDF for particles in a gas, we can use the definitions listed in 2.1.1 to find a general expression. First, we can find the quantity $\langle v^2 \rangle$ by (notice the bounds of integration from eq. 2.6)

$$\langle v^2 \rangle = \int_0^\infty v^2 f(v) dv \tag{2.8}$$

where $f(v)$ is the Maxwell-Boltzmann distribution. Plugging in $f(v)$ and rearranging yields

$$\frac{2}{\sqrt{\pi}} a^{3/2} \int_0^\infty v^4 e^{-av^2} dv.$$

Where we use $a = m/2k_BT$. Integrating by parts with $u = v^3, du = 3v^2 dv$ and $dw = ve^{-av^2} dv, w = -\frac{1}{2a} e^{-av^2}$, we have

$$\frac{2}{\sqrt{\pi}} a^{3/2} \left[-\frac{v^3}{2a} e^{-av^2} \Big|_{-\infty}^{\infty} + \frac{3}{2a} \int_0^\infty v^2 e^{-av^2} dv \right] \tag{2.9}$$

The first term in the brackets is zero since the exponential decays to zero at infinity. To find the integral on the right, we could use integration by parts again, but instead we can use Feynman's trick for a quicker solution. By properties of the Gaussian integral [3],

$$\int_0^\infty e^{-av^2} dv = \frac{1}{2} \sqrt{\frac{\pi}{a}}.$$

Comparing this property with the integral in eq. 2.8, we can observe the following clever relation

$$\int_0^\infty v^2 e^{-av^2} dv = -\frac{d}{da} \int_0^\infty e^{-av^2} dv$$

[3] I won't prove it here, but the derivation is very interesting and you should definitely check it out!

where we take a derivative with respect to a, since the integral is a function of a [4]. Thus, the expression in eq. 2.8 becomes

$$\begin{aligned}\langle v^2 \rangle &= \frac{4}{\sqrt{\pi}} a^{3/2} \left[\frac{3}{2a} \left(-\frac{d}{da} \frac{1}{2} \sqrt{\frac{\pi}{a}} \right) \right] \\ &= \frac{4}{\sqrt{\pi}} a^{3/2} \left[\frac{3}{2a} \left(\frac{1}{4} \frac{\sqrt{\pi}}{a^{3/2}} \right) \right] \\ &= \frac{3}{2a}. \end{aligned} \tag{2.10}$$

The RMS speed is the square root of this value, or

$$\begin{aligned} v_{rms} &= \sqrt{\langle v^2 \rangle} \\ &= \sqrt{\frac{3}{2a}} \\ &= \sqrt{\frac{3k_B T}{m}} \end{aligned} \tag{2.11}$$

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[4] v is a dummy variable, and is integrated out